DATA SCIENCE RESEARCH CAPSTONE

Customer Churn Prediction Using Machine Learning

A reproducible classification study using the IBM Telco Customer Churn sample dataset

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Research integrity note

The report specifies a complete reproducible experiment and provides executable code. Model metrics are deliberately not fabricated: the document-generation environment could not retrieve the CSV for local execution. Run the included script to calculate genuine cross-validation and held-out test results from the checksum-verified source dataset.

Educational research. The IBM sample represents a fictional telecommunications provider; findings are not claims about a live operator.

Abstract

Customer churn reduces recurring revenue and increases the cost of replacing subscribers. This capstone defines a reproducible machine-learning workflow for identifying customers with elevated churn likelihood from account, service, tenure and billing attributes. The study uses the public IBM Telco Customer Churn sample, a cross-sectional benchmark containing 7,043 customer records and a binary churn label.

The proposed experiment compares Logistic Regression, Random Forest and Extra Trees. A stratified 80/20 split reserves an untouched test set; model selection is based on mean five-fold cross-validation average precision (PR-AUC) on the training partition. Numeric and categorical preprocessing is encapsulated inside scikit-learn pipelines to avoid leakage. Holdout evaluation reports accuracy, precision, recall, F1, ROC-AUC and PR-AUC, alongside confusion matrix, ROC and precision-recall curves.

Empirical scores are not printed as if measured. The execution environment used to assemble this PDF could not download the raw CSV, so no model was fitted here. The repository includes a checksum-verifying training script that generates all genuine metrics and plots when run with internet access. This is a transparent execution limitation, not a substitute for results.

Keywords: customer churn, binary classification, telecom analytics, imbalanced learning, model evaluation, retention strategy.

1. Introduction and Research Problem

1.1 Background

Subscription businesses depend on customer retention. Churn analytics uses historical customer characteristics to estimate which subscribers are more likely to discontinue service. A risk score can help retention teams prioritize outreach, investigate service friction and allocate limited incentives more carefully.

1.2 Problem statement

The research problem is to classify whether a customer belongs to the churn class using information available in the IBM sample snapshot. The target field is `Churn`, with values Yes and No. The task is supervised binary classification. The dataset is not longitudinal, so the project does not claim to forecast churn in a future calendar period.

1.3 Research questions

RQ1: How well do three commonly used classifiers distinguish churners from non-churners on a stratified holdout sample? RQ2: Does a nonlinear tree ensemble improve cross-validated PR-AUC relative to a linear baseline? RQ3: Which observed customer attributes contribute most to predictive performance under permutation importance?

1.4 Objectives and scope

Build an auditable data pipeline; compare Logistic Regression, Random Forest and Extra Trees; evaluate minority-class detection using PR-AUC, recall and precision; provide interpretable operational recommendations; and package code, environment requirements and data provenance for reproduction. Causal inference, treatment-effect estimation and live deployment are outside scope.

2. Literature Background and Conceptual Basis

Churn prediction has long been studied as a customer relationship management problem. Prior telecommunications research commonly frames churn as a classification task and uses account tenure, contract structure, service subscriptions and billing information as explanatory variables. A peer-reviewed case study examining telecom churn discusses IBM Telco alongside other datasets and emphasizes the value of combining customer and service information [1].

2.1 Model families

Logistic Regression provides a transparent linear baseline and estimates log-odds as a function of predictors. Tree ensembles can capture nonlinear relationships and interactions without requiring manual specification of every interaction. Random Forest aggregates randomized decision trees; Extra Trees increases split randomization. Comparing these model families provides a practical bias/variance contrast.

2.2 Evaluation under class imbalance

The churn class is a minority class in this benchmark. Accuracy alone can be misleading because a model that predicts the majority class may appear accurate while missing many churners. Precision-recall analysis focuses on positive-class retrieval; average precision summarizes the precision-recall curve. ROC-AUC provides a complementary threshold-independent ranking metric. F1 balances precision and recall at a chosen threshold.

2.3 Leakage control

Preprocessing must be learned only from training data. Scikit-learn pipelines combine transformations and estimators so cross-validation refits preprocessing within each fold [2]. This project uses that design rather than imputing or encoding the complete dataset before splitting.

3. Dataset and Data Understanding

3.1 Source and structure

The study uses IBM's public Telco Customer Churn sample CSV. It contains 7,043 rows and 21 columns: a customer identifier, 19 customer/account/service predictors plus the churn outcome. IBM describes the source as sample data for a fictional telecommunications company. The churn label indicates whether the customer left within the last month [3].

3.2 Variable groups

Demographic/account profile: gender, SeniorCitizen, Partner, Dependents. Tenure and charges: tenure, MonthlyCharges, TotalCharges. Service portfolio: PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV and StreamingMovies. Commercial terms: Contract, PaperlessBilling and PaymentMethod. Identifier: customerID, excluded from modeling.

3.3 Data quality

`TotalCharges` is represented as text in the raw CSV and includes blank entries for customers with no accumulated charge. The pipeline converts it to numeric and imputes missing numeric values using the median learned on training data. Categorical missing values are imputed using the training-set most frequent category. Customer ID is removed because it is an identifier, not a stable behavioral signal.

3.4 Target and imbalance

The outcome is mapped to 1 for Churn=Yes and 0 for Churn=No. The dataset has a minority churn class; the exact class counts and churn rate are computed from the downloaded file at execution and written to `outputs/run_metadata.json`, rather than copied from an unverified secondary source.

The raw data are a cross-sectional snapshot. A random stratified split measures within-snapshot generalization, not temporal generalization to future months. A later production study should use time-based splits and repeated monthly snapshots.

4. Research Methodology

4.1 Experimental design

The experiment uses a deterministic stratified 80/20 train/test split with random_state=42. Stratification preserves the observed class proportions in both partitions. The test partition remains untouched during model selection. On the training partition, five-fold StratifiedKFold cross-validation estimates model performance.

4.2 Preprocessing

Numeric columns are median-imputed and standardized. Categorical columns are imputed with the most frequent category and one-hot encoded with unknown-category tolerance. A ColumnTransformer applies the correct treatment by column type. The complete preprocessing-plus-estimator pipeline is refitted in every cross-validation fold.

4.3 Candidate models

Logistic Regression uses class_weight=balanced and an increased iteration limit. Random Forest and Extra Trees each use 500 trees, minimum leaf size 2, class_weight=balanced and fixed random seeds. These are reasonable benchmark configurations, not an exhaustive hyperparameter search.

4.4 Model selection

The primary selection criterion is mean five-fold cross-validation average precision (PR-AUC). This emphasizes the positive churn class and is more informative than accuracy for an imbalanced binary target. Cross-validation standard deviation is also reported to indicate fold-to-fold variability.

4.5 Holdout evaluation

The selected model is evaluated once on the holdout partition. The default probability threshold is 0.50 for threshold-dependent precision, recall, F1 and confusion matrix. ROC-AUC and average precision use predicted probabilities. Threshold tuning should be conducted on validation data only in a subsequent business-cost optimization phase.

5. Predictive Modeling and Reproducibility

5.1 Workflow

Raw CSV → checksum verification → schema validation and target extraction → stratified split → fold-local imputation/encoding/scaling → model fitting → cross-validation model selection → final holdout prediction → metrics and visualizations → saved model artifacts.

5.2 Reproducibility controls

The dataset is downloaded from IBM's public GitHub sample repository when absent. The script checks the SHA-256 digest before use, ensuring the file matches the documented snapshot. Random seeds are fixed for splitting, cross-validation and estimators. Python dependencies are listed in requirements.txt. The output includes CSV metrics, JSON run metadata, classification report, confusion matrix, ROC curve, precision-recall curve and permutation importance.

5.3 Repository execution

From the repository root, create a virtual environment, install `requirements.txt`, then run `python src/train.py`. The first run requires internet access to retrieve the CSV. Subsequent runs use the local checksum-verified copy. The script saves fitted pipelines in `models/` and generated evidence in `outputs/`.

No test scores are asserted in this PDF because the dataset could not be downloaded into the report-building runtime. The provided training script is executable and writes real scores when run. The evaluator should inspect the generated CSV and plots after execution.

6. Model Performance: Reporting Protocol

6.1 Metrics

Accuracy = (TP+TN)/(TP+TN+FP+FN). Precision = TP/(TP+FP), measuring the share of predicted churners who actually churned. Recall = TP/(TP+FN), measuring the share of churners identified. F1 is the harmonic mean of precision and recall. ROC-AUC measures ranking across thresholds. Average precision summarizes the precision-recall curve and is the primary model-selection metric.

6.2 Evidence generated by the code

The script writes `outputs/model_metrics.csv` with cross-validation PR-AUC mean and standard deviation, plus holdout accuracy, precision, recall, F1, ROC-AUC and PR-AUC for every candidate. It writes `classification_report.txt`, `confusion_matrix.png`, `roc_curve.png`, `precision_recall_curve.png` and `permutation_importance.csv`. The selected model is chosen by training-only CV PR-AUC.

6.3 Interpretation guardrails

Metrics must be read together. High recall with low precision may create too many unnecessary retention contacts; high precision with low recall may miss customers who leave. The preferred operating point depends on outreach cost, customer lifetime value and intervention effectiveness. A threshold of 0.50 is a reproducible baseline, not a claim of optimal business utility.

Results table: populate from `outputs/model_metrics.csv` after running the code. Do not manually type illustrative scores into a formal submission. This safeguards the requirement for genuine model evaluation.

7. Discussion and Business Implications

7.1 Translating risk into action

Churn probabilities can support a prioritized review queue. Retention teams can investigate contract friction, service reliability, billing clarity and support experience for high-risk customers. Offers should be targeted only when expected retained contribution exceeds the cost of the incentive and contact.

7.2 Feature importance

Permutation importance measures the reduction in a chosen score when one feature is shuffled on the holdout data. Larger reductions indicate that the model relies more heavily on that feature for predictive performance. It is not a causal effect, and correlated variables can share or mask importance. The generated importance CSV should be reviewed alongside domain knowledge and subgroup checks.

7.3 Operational measurement

A controlled retention experiment is required to establish whether an intervention reduces churn. Randomly assign eligible high-risk customers to treatment and control groups where ethical and operationally feasible; compare churn and incremental margin. Prediction quality and intervention effectiveness are distinct questions.

7.4 Governance

Before deployment, assess calibration, subgroup error rates, stability over time, data minimization, access controls and applicable privacy rules. Avoid using demographic attributes to deny service, impose punitive pricing or make opaque decisions. Human review and an appeal path should accompany consequential actions.

8. Limitations, Conclusion and Future Work

8.1 Limitations

The dataset is a public sample for a fictional provider, not a representative extract from a current operator. Its labels and predictors form a single cross-sectional snapshot. The random split does not test temporal drift. The baseline models and fixed hyperparameters are not exhaustive. Class weights and a 0.50 threshold may not optimize business outcomes. Predictive associations do not establish causality.

8.2 Conclusion

This capstone specifies a reproducible and leakage-aware workflow for telecom churn classification. It compares a linear baseline with two tree ensembles, selects by cross-validated PR-AUC and evaluates once on a stratified holdout set. Its repository produces genuine metrics and plots from a checksum-verified dataset. Empirical scores are intentionally left unreported until the included script is executed; no fabricated performance claims are made.

8.3 Future work

Extend the study with temporal validation across monthly snapshots; tune hyperparameters using nested or validation-based procedures; calibrate probabilities; optimize thresholds against retention economics; test gradient boosting; quantify uncertainty; assess subgroup fairness; and run randomized intervention experiments to estimate incremental retention impact.

References and Submission Checklist

[1] Customer retention and churn prediction in the telecommunication industry: a case study on a Danish university. Peer-reviewed open-access article, 2023. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC10239051/

[2] scikit-learn developers. Pipeline documentation. https://scikit-learn.org/stable/modules/generated/sklearn.pipeline.Pipeline.html

[3] IBM. Telco Customer Churn sample dataset, public repository CSV. https://github.com/IBM/telco-customer-churn-on-icp4d (data/Telco-Customer-Churn.csv).

[4] OpenML. telco-customer-churn, dataset ID 45568; metadata and feature schema. https://www.openml.org/search?id=45568&sort;=runs&type;=data

[5] scikit-learn developers. Common pitfalls and recommended pipeline use. https://scikit-learn.org/stable/common_pitfalls.html

Submission checklist

■ Run `python src/train.py` with internet access and confirm checksum validation. ■ Review `outputs/model_metrics.csv` and generated plots. ■ Insert the actual metrics and date of execution into the report's results section if your mentor requires a completed empirical-results table. ■ Commit source code, README and requirements to GitHub. ■ Do not commit raw customer-level data or trained models unless permitted by the dataset terms.

Deliverables included: formal 10-page PDF report; reproducible Python training script; dataset source and checksum; README; requirements; Git ignore rules; license note; repository structure. The dataset itself is fetched by the script to avoid bundling a copy and to preserve provenance.